



Artificial intelligence applications in adaptive radiotherapy—a narrative review

Zeja F. Chen^{1^}, Benjamin Larocque^{1^}, Jade Wang¹, David Chen^{2^}, Wei Liu³, Herve H. F. Choi^{4^}, Srinivas Raman^{3^}

¹School of Medicine, Queen's University, Kingston, ON, Canada; ²Temerty Faculty of Medicine, University of Toronto, Toronto, ON, Canada;

³Division of Radiation Oncology, Department of Oncology, BC Cancer, University of British Columbia, Vancouver, BC, Canada; ⁴Department of Medical Physics, BC Cancer, Vancouver, BC, Canada

Contributions: (I) Conception and design: ZF Chen, S Raman; (II) Administrative support: S Raman; (III) Provision of study materials or patients: S Raman; (IV) Collection and assembly of data: ZF Chen, B Larocque, J Wang; (V) Data analysis and interpretation: ZF Chen, B Larocque, J Wang; (VI) Manuscript writing: All authors; (VII) Final approval of manuscript: All authors.

Correspondence to: Srinivas Raman, MD, FRCSC. Division of Radiation Oncology, Department of Oncology, BC Cancer, University of British Columbia, 600 10th Ave W, Vancouver, BC, V5Z 4E6, Canada. Email: srinivas.raman@bccancer.bc.ca.

Background and Objective: Online adaptive radiotherapy (ART) is an emerging tool to adapt radiotherapy plans to interfraction changes during the treatment course, reducing dose to organs at risk (OARs) while ensuring adequate dose to the tumor volume. Artificial intelligence (AI) can be a promising tool to support online ART techniques given that daily treatment changes can be resource- and time-intensive. With increasing integration of AI into radiation treatments, this literature review summarizes current applications and promising future directions of AI for online ART.

Methods: A comprehensive search of databases (EMBASE, MEDLINE, PsycInfo, PubMed, CINAHL, CENTRAL, Scopus, and Web of Science, 2015–2025) was performed. Examples of search terms include “artificial intelligence”, “synthetic image generation”, “autosegmentation”, and “adaptive radiotherapy”. Selected studies were peer-reviewed studies that describe, validate, or evaluate an AI tool for any online ART-related function in the domains of decision support, imaging, contouring, planning, and quality assurance (QA).

Key Content and Findings: AI applications for online ART can be divided broadly into three domains: image registration, auto-contouring of tumor volumes and OARs, and ART decision support through dose and anatomical change prediction. New deep-learning based synthetic computed tomography (CT) generation using architectures like generative adversarial networks have enabled magnetic resonance imaging (MRI)-guided adaptive workflows. AI-driven deformable image registration (DIR) has also been shown to demonstrate alignment between planning and on-treatment imaging. Similarly, AI-driven enhancement of cone-beam CTs has improved their resolution, improving spatial information valuable for segmentation of tumor volumes. Auto-segmentation using U-Nets, ResNets, and one-shot learning has substantially reduced contouring time and approaches expert performance for many organs at risk (OARs), though challenges persist for small structures and evolving gross tumor volume (GTV)/clinical target volume (CTV) volumes. AI-based dose prediction and personalized plan optimization demonstrate clinically meaningful reductions in OAR dose, and early commercial systems such as Varian Ethos show promising real-world feasibility. Decision-support systems leverage these AI-workflows, using predictive modeling to guide daily dose adjustments.

Conclusions: AI has the potential to transform ART, automating labour-intensive tasks while still achieving high-quality plans. To support safe clinical adoption, future work should emphasize multi-

[^] ORCID: Zeja F. Chen, 0000-0002-4286-7871; Benjamin Larocque, 0009-0002-6142-3632; David Chen, 0000-0003-3178-519X; Herve H. F. Choi, 0009-0000-4226-3210; Srinivas Raman, 0000-0001-5688-9628.

institutional validation, standardized benchmarking and contouring guidelines, and development of transparent, uncertainty-aware models.

Keywords: Adaptive radiotherapy (ART); artificial intelligence (AI); radiation oncology; cancer; oncology

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Introduction

Radiotherapy is a cornerstone of cancer treatment, and over 50% of cancer patients will require treatment during their lifetime. In the last two decades, technological improvements in radiotherapy planning and image guidance have increased the precision and accuracy of treatment, and these improvements have been associated with decreased toxicities.

Adaptive radiotherapy (ART) allows radiation oncologists to tailor treatment in response to changes in patient anatomy or tumor position that occur during the course of radiotherapy, accounting for interfraction motion. ART can be carried out using cone beam computed tomography (CBCT)-based machines or magnetic resonance (MR)-Linac guided machines. After an initial simulation planning scan on which an initial radiation plan is created, future treatments involve daily CBCTs or magnetic resonance imagings (MRIs) to assess patient anatomy variation from day to day. The initial simulation plan is then adapted using a variety of techniques to account for variations in anatomy.

ART encompasses multiple levels of adaptiveness that differ substantially in their technical requirements, response times, and degree of human oversight. Offline ART involves plan modification between treatment fractions and allows hours to days for image review, contouring, planning, and quality assurance. Online ART, which is the primary focus of this review, requires that these processes be completed at the treatment console within a single treatment session, typically within minutes, while the patient remains on the treatment couch. This setting imposes stringent constraints on computational efficiency, robustness, and safety, and necessitates human-in-the-loop verification of AI-generated outputs. Real-time ART, in contrast, would require sub-second system response and minimal human intervention, and remains largely investigational at present. Given the fundamentally different operational, regulatory, and clinical requirements of these paradigms, this review intentionally focuses on AI applications that have been developed or

evaluated specifically for online ART, where AI has the most immediate and clinically meaningful impact (1-3).

While online ART may offer clinical benefits for patients, implementing it consistently is resource intensive (1,4). It demands frequent re-imaging and re-contouring of targets and organs at risk (OARs), as well as treatment re-planning and quality assurance within tight timeframes. For context, delineation of target volumes and OARs may take a few hours in a head-and-neck cancer patient and there can be significant interobserver variability (5-7). Artificial intelligence (AI) is increasingly being investigated as a promising solution to these limitations. Over the last few years, deep-learning models have been developed to automate key parts of the online ART process. AI can be integrated in the workflow through (I) automatic generation of contours for target volumes and OARs; (II) creating synthetic CT scans for MR-based ART; (III) dose calculations from generated contours and treatment plans; and (IV) clinical decision-making on treatment for interfraction variability.

This paper aims to characterize the landscape of AI applications in ART, particularly in treatment decision support, segmentation of OARs and target volumes, and dose optimization. We also discuss key gaps in literature and the challenges in translating AI into practice. We present this article in accordance with the Narrative Review reporting checklist (available at <https://tcr.amegroups.com/article/view/10.21037/tcr-2025-1-2641/rc>).

Methods

We identified relevant studies and potential applications of AI in online ART by conducting a comprehensive but non-systematic search of academic databases: EMBASE, MEDLINE, PsycInfo, PubMed, CINAHL, CENTRAL, Scopus, and Web of Science (*Table 1*). Examples of search terms include “artificial intelligence”, “synthetic image generation”, “autosegmentation”, and “adaptive

Table 1 Search strategy summary

Items	Specification
Date of search	June 26, 2025
Databases and other sources searched	EMBASE, MEDLINE, PsycInfo, PubMed, CINAHL, CENTRAL, Scopus, and Web of Science
Search terms used	(artificial intelligence or machine learning or deep learning) AND (adaptive radiotherapy or adaptive rad* or adaptive techniques or online adaptive or real-time adaptive or offline adaptive) AND (contouring or segmentation or image generation or synthetic image or image registration or dose or dosimetry or dose support or dose prediction or decision support or workflow)
Timeframe	January 1, 2015 to January 1, 2025
Inclusion and exclusion criteria	Peer-reviewed studies written in English that describe, validate, or evaluate an AI algorithm or tool for any ART-related function (decision, imaging, contouring, planning, QA). Studies excluded were non-English, applied to conventional ART without adaptive elements, and did not have AI use in their workflows
Selection process	Studies were independently reviewed by three authors (Z.F.C., B.L., X.W.) and relevant contents were extracted for thematic analysis

AI, artificial intelligence; ART, adaptive radiotherapy; QA, quality assurance.

radiotherapy”. Studies were searched from 2015–2025 in English to identify recent innovations. Inclusion criteria for selected studies were peer-reviewed studies that describe, validate, or evaluate an AI tool for any ART-related function in the domains of decision support, imaging, contouring, planning, and QA. We also cross-referenced bibliographies of retrieved articles and drew on clinical and AI knowledge within the author team to ensure coverage of emerging and notable studies.

Discussion

We categorize and describe applications of AI in ART in three key areas: (I) image registration; (II) auto-contouring; and (III) decision support (including dose planning and anatomical change prediction tools). *Figure 1* provides an overview summarizing current techniques integrating AI into the online ART workflow.

AI-based synthetic image generation

Conventionally, image-guided radiotherapy has relied on kilovoltage or megavoltage X-ray imaging. MRI has become a popular modality for guiding radiotherapy due to its superior soft tissue contrast and lack of ionizing radiation in comparison to CT or X-ray (8). However, traditionally a CT image is required for external beam planning (9), as the voxel intensity of CT images directly correlates to the electron density or physical density of the tissue, which is

required for dose calculation. For an online ART workflow where only MRI images are available during treatment, a synthetic CT scan (sCT) is created. sCTs are images derived from other modalities like MRI through an estimation of tissue electron density and attenuation. Although traditionally, producing an sCT requires a reference CT scan for calibration, modern workflows are now being developed for CBCT or MRI-only approaches to create synthetic CTs without prior references. For example, sCT generation from low-field MRI, lightweight systems under 1 Tesla, have been integrated with linear accelerators for ART. sCT generation reduces the need for daily CT scans, improves resource stewardship and reduces registration error (10).

One of the first applications of deep learning for the generation of sCT from low-field MRI images for the purpose of ART has been reported by Cusumano *et al.* in 2020 (11). The study used a conditional Generative Adversarial Network (cGAN) in which two neural networks are trained in competition with each other. One network was tasked with creating the sCT (the generator) while the other evaluated the image quality (the discriminator), providing feedback to the generator network. Of the 9,950 reference CT and MRI images available from 80 patients in the training cohort, the cGAN was trained on 4,848 (2,681 pelvis, 2,167 abdomen) images that passed quality control by a radiation oncologist. The generator used a U-Net architecture with an encoder path for feature compression and a decoder path for image reconstruction. A PatchGAN

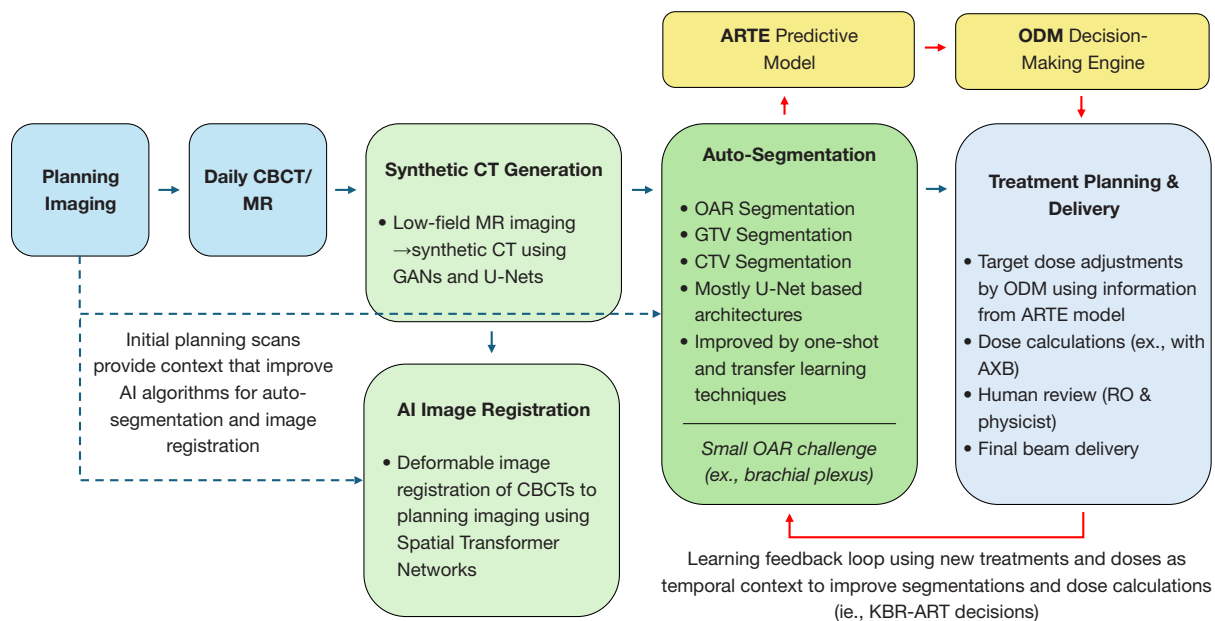


Figure 1 Areas integrating AI into the adaptive radiotherapy workflow. AI, artificial intelligence; ARTE, Artificial RT Environment; AXB, Acuros XB; CBCT, cone beam computed tomography; CT, computed tomography; CTV, clinical tumor volume; GANs, generative adversarial networks; GTV, gross tumor volume; KBR-ART, Knowledge-Based Response-Adaptive Radiotherapy; MR, magnetic resonance; OAR, organs at risk; ODM, optimal decision making; RO, radiation oncologist; RT, radiotherapy.

architecture was used as the discriminator after image reconstruction and evaluated image realism in 70×70 voxel patches. The model was trained to minimize the error rate of the discriminator on generated images. After training was complete, the mean time for the neural network to generate a sCT from MRI was 175 ± 43 seconds for pelvic cases and 110 ± 40 seconds for abdominal cases. Notably, the 2–3 minutes required to generate a set of sCT images is much faster than the time frame required compared to atlas-based (80 minutes) or non-deep learning machine learning methods (62 minutes) (12,13). The sCT dose calculations generated by the cGAN were within clinically accepted dosimetric tolerances compared to actual CT-based planning, as measured by mean gamma passing rates in 40 patients. However, a notable limitation of the study was exclusion of poor image quality scans, limiting external validity.

While these deep learning based sCT generation methods significantly reduced time of generation compared to atlas-based methods, further improvements are needed to support online ART. In online ART, sCT generation must be fast to support same-day dose recalculation and plan adaptation. Additionally, imaging quality restrictions make it difficult to allow for robust generation especially in

online ART, where sCT synthesis must be robust to variable anatomy, and tightly integrated with automated dose engines; otherwise, errors in sCT mapping can directly bias daily adaptation decisions.

Hsu *et al.* report an improved architecture for generating sCT from low-weight MRI in prostate cancer patients (14). Unlike Cusumano *et al.* and similar studies that employed a 2D cGAN architecture, Hsu *et al.* [2022] elected to use a multi-planar approach capable of processing sagittal, and coronal MRI slices to generate a volumetric sCT from MRI data in any orthogonal direction. A U-Net and PatchGAN architecture were used as a generator and discriminator, creating the GAN framework. However, the generator was provided with multidimensional data in the form of three MRI sets (one per direction). Each was processed into a synthetic CT image set, which were then averaged to obtain a final CT volume. The multi-planar approach offers a key potential advantage in that it reduces discontinuities common in two-dimensional (2D) models, producing more accurate electron density information. Dose differences were small and clinically acceptable, typically within 1% to 2% for planning target volumes (PTVs) and OARs. Although this technique did not yield any major improvements in dose, this study's methodology

included unclear imaging and artifacts, which was excluded in Cusumano *et al.*, improving on external validity. Despite increased complexity, sCT generation time was reduced to approximately 12 seconds using a high-performance graphics processing unit (GPU).

Similarly, Zhao *et al.* describe a platform for generation of sCTs from CBCT images specifically to enable online CBCT-based ART, where poor image quality otherwise limits reliable dose recalculation and re-planning (15). Using a similar cGAN architecture, 40 CBCTs taken from rectal cancer patients were converted to synthetic CT scans on-treatment. The study reported a structural similarity index of 0.81 ± 0.03 in the sCT group compared to regular CT imaging, demonstrating far greater similarity to conventional planning CT than raw CBCT imaging (0.44 ± 0.07). Crucially to adaptive workflows, autosegmentation of clinical target volume (CTV) and OARs such as the bladder demonstrated 0.93 and 0.94 concordance between sCT and CT imaging, and dose plans generated on sCT and CT imaging differed at Dmean, D99, D5 and D95 by only ~50 cGy, where CBCT dose differences could vary up to 350 cGy. These findings are important in online ART, where CBCTs are often the only daily image. Generating accurate sCTs from CBCT imaging improves contour reliability and plan fidelity, reducing erroneous adaptive decision making. This study demonstrates that in addition to MR-only workflows, synthetic CT generation can be a useful step in CBCT-based adaptive workflows.

Current MR-guided platforms like the Elekta Unity and ViewRay MRIdian generate electron density information to create sCTs via traditional methods like bulk density assignment, atlas-based methods, or deformable registration of a prior planning CT to daily MRI. Comparison of sCT generation time between AI models and commercial systems will be necessary to understand the value of AI methods compared to traditional methods and allow for translation into industry or clinical practice. Nevertheless, with growing improvements to AI methods that quickly and accurately generate sCTs, there is potential for AI to be a foundational element of online ART systems.

Auto-contouring with AI

Autosegmentation refers to the automatic delineation of anatomical structures and tumors on medical imaging. These algorithms are trained on large, annotated datasets to learn spatial and textural features to identify both

normal organs and tumors. Once trained, they can generate segmentations within minutes, dramatically reducing the time required compared to manual contouring in complex cases, which can take hours. These workflows have already been integrated into diagnostic radiology and radiation oncology, where segmentation techniques have been successfully applied to tasks such as liver lesion detection, lung volume segmentation, and brain tumor delineation (16-18). These successes paved the way for their adaptation to radiotherapy. However, ART poses unique challenges to segmentation, as it requires repeated delineation across treatment sessions, needing reliable and fast contouring tools for the clinical workflow, often on lower quality imaging, although the quality of on-board CT and MRI imaging have significantly improved in recent years. This section aims to discuss methods currently addressing challenges faced in segmentation in the adaptive setting.

OARs segmentation

OAR segmentation has been the most widely studied and clinically implemented application of AI contouring in ART (1,19). OAR segmentation has achieved strong performance in the context of planning CT and high-quality diagnostic MRI imaging. However, a major challenge specific to ART is the segmentation of OARs from low image quality on-treatment imaging.

Early segmentation techniques in the context of online ART used atlas-base approaches; however, with the recent advent of deep-learning, new segmentation algorithms are being developed for OARs, with the majority using U-Net models and convolutional neural networks (CNNs) (20,21). CNNs are a type of deep learning algorithm which uses a convolutional filter to identify lines, curves, or textures from images, combining them to make predictions for recognition of more complex objects. U-Net models, a type of CNN, combines this information with an encoder-decoder architecture to predict and construct a contour or segmentation. The Dice similarity coefficient (DSC) is one of the most commonly used metrics to evaluate the performance of U-Net models, and measures the overlap of the generated contour, A, with an established ground truth segmentation, B (22). This is scored from 0 to 1, where 1 represents perfect agreement, and is represented using the equation:

$$\text{Dice}(A, B) = \frac{2 \times |A \cap B|}{|A| + |B|} \quad [1]$$

U-Net models have achieved strong results in adaptive treatments of some treatment sites, with median DSCs across most sites over 0.75 compared to the ground truth labels. In direct comparison to traditional atlas-based methods, U-Net based models have improved OAR segmentation in multiple sites, including head and neck and gastrointestinal (GI) structures (23,24). *Table 2* shows OARs and contouring accuracy in adaptive plans in head and neck, pelvic, and thoracic sites (25-32).

However, a remaining criticism of the U-Net architecture is its difficulty with segmenting small, low-contrast OARs. In online adaptive settings, low on-treatment image quality reduces signal strength of smaller OARs, leading to consistently lower concordance using AI-based autosegmentation techniques (33). Although there is no volumetric size threshold below which these algorithms fail, OARs where deep learning methods underperform include the optic chiasm, pituitary gland, or submandibular ducts, with reported DSCs as discordant as 0.37 (34). Treatment sites like head and neck often require segmentation of multiple small, complex, and low-contrast OARs on low-quality on-treatment imaging, leading to difficulty in optimizing DL networks. To help the U-Net encapsulate a higher resolution needed for smaller, complicated anatomy, Gao *et al.* demonstrated implemented a two-stage deep neural network, using a small-organ localization sub-network. This sub-network implements a separate U-Net and autoencoder trained on a dataset specific to small OARs on CT imaging. Segmentations done by the sub-network are used as regions of interest for small OARs, and then mapped to a regular U-Net segmentation to create the final segmentation (35). This method outperformed traditional U-Nets, with a DSC of 0.74 compared to results of 0.37 to 0.71 achieved by other algorithms tested. This technique demonstrates potential improvements in the adaptive workflow, increasing segmentation accuracy of small structures on low-quality imaging.

Later, a stratified OAR segmentation (sOARs) AI model developed by Guo *et al.* leveraged this work to detect and classify OARs by spatial complexity, which then enters an algorithm to identify the optimal neural network to utilize (U-Net or modified U-Net, optimized for smaller OARs). This model outperformed conventional segmentation by 5% in DSC, with 98% of sOARs requiring only minor revisions by radiation oncologists (36). Another option studied has been to enhance the quality of the imaging itself. Chen *et al.* demonstrated in 40 head and neck patients that a U-Net architecture trained on planning

CT as well as first fraction cone beam CT to create an enhanced cone beam CT image can improve delineation of head and neck OARs (28).

Commercial platforms have integrated AI algorithms and deployed this in real world settings, approaching expert-level contours in multiple treatment sites (37,38). Contouring systems include Organs RT by Siemens Healthineers, Contour+ by MVision, Annotate by TheraPanacea, and AutoContour by RadFormation (39). A comparison of some of these systems on breast, head and neck, lung, and prostate patients showed strong performance compared to existing literature, with DSCs between 0.82 and 0.88 (40). However, these algorithms have similarly demonstrated struggles segmenting lower resolution structures. For example, in the CNS site, the MVision system achieved a median DSC of 0.74, with a DSC of 0.44 for smaller structures like the pituitary gland (41). Integration of novel AI algorithms can improve small OAR segmentation and utility in online adaptive workflows.

Despite its popularity of use, the DSC may not predict clinical adequacy of contours. The DSC has limitations in identifying proximity to critical structures and cannot differentiate between random and systematic errors. Clinical validation is essential; however, the field remains constrained by few large-scale, multi-institutional validation studies. Part of the difficulty of this is variability in contouring guidelines between centers, contributing to inter- and intra-observer variability in datasets. Adoption of established international contouring guidelines can ensure reproducibility and generalizability, producing better quality datasets amenable for widespread clinical correlation. For example, the American Association of Physicists plans to form guidelines on adopting automatic segmentations, part of which includes providing standardized recommendations for validation, acceptance testing, commissioning, and maintenance of these tools (42).

Gross tumor volume (GTV) and CTV segmentation

Compared to OAR segmentation, automating GTV and CTV delineation is a more challenging problem. Unlike OARs, which can remain relatively stable throughout treatment, both GTVs and CTVs evolve over time due to tumor shrinkage, treatment response, and anatomical changes. Additionally, CTV segmentation poses additional complexity because of clinical judgement regarding microscopic disease and subclinical nodal involvement, which cannot be directly visualized easily on lower quality daily cone-beam CT and MRI techniques, requiring AI

Table 2 A summary of Dice similarity coefficients of organs at risk in head and neck, genitourinary, gynecologic, and thoracic sites

Site	Study	Number of patients	Neural network architecture	Imaging modality	Organ at risk	Dice similarity coefficient
Head and neck	Kakkos <i>et al.</i> (25)	113	AttentionU-Net	Cone-beam CT	L/R parotid gland	0.83
	Liang <i>et al.</i> (26)	239	CycleGAN	Cone-beam CT	Spinal cord	0.88
					L SMG	0.79
					R SMG	0.77
					L/R masseter	0.89
					Mandible	0.89
					Larynx	0.86
					Esophagus	0.81
					Constrictor	0.75
					Oral cavity	0.94
					Brainstem	0.91
					L brachial plexus	0.66
					R brachial plexus	0.71
	Ma <i>et al.</i> (27)	37	U-Net	Cone-beam CT	Mandible	0.93
					L parotid	0.86
					R parotid	0.89
					L SMG	0.81
					R SMG	0.77
					L masseter	0.91
					R masseter	0.89
					Chen <i>et al.</i> (28)	55
	R parotid	0.84				
	L SMG	0.75				
	R SMG	0.77				
	Larynx	0.84				
	Brainstem	0.80				
	Spinal cord	0.88				
Genitourinary	Nachbar <i>et al.</i> (29)	47	Annotate by TheraPanacea	Low-field MRI		
					Bladder	0.97
					Penile bulb	0.74
					Prostate	0.87
					Rectum	0.90
					Seminal vesicles	0.78

Table 2 (*continued*)

Table 2 (continued)

Site	Study	Number of patients	Neural network architecture	Imaging modality	Organ at risk	Dice similarity coefficient
Gynecologic			U-Net	Low-field MRI	Sacrum	0.90
					L pelvis	0.92
					R pelvis	0.91
					L femur	0.94
					R femur	0.93
	Chen <i>et al.</i> (30)	26	U-Net	Low-field MRI	Bladder	0.91
	Breto <i>et al.</i> (31)	15	R-CNN	Low-field MRI	Rectum	0.82
					Femoral heads	0.92
					Mesorectum	0.69
					Rectum	0.88
					Uterus	0.69
					Vagina	0.18
					Parametrium	0.58
					Sigmoid	0.69
					Femur	0.76
					Bladder	0.75
Thoracic	Wang <i>et al.</i> (32)	9	U-Net	Low-field MRI	L/R lung	0.98
					Heart	0.94
					Spinal cord	0.95

CNN, convolutional neural network; CT, computed tomography; L, left; MRI, magnetic resonance imaging; R, right; SMG, submandibular gland.

models capable of recognizing tumor regression over time on on-treatment image quality scans.

Historically, auto-segmentation of targets in the online ART workflow has been through deformable image registration (DIR)-based methods. Contours determined by experts on planning CTs are deformed to a CBCT image using deformable vector fields (DVF) (43). Recently, deep learning (DL)-based DIR methods have shown excellent performance compared to traditional DIR methods. These models include Spatial Transformer Networks and U-Nets that can directly predict the DVF needed to propagate contours from planning CTs to cone-based plans (44,45). However, these techniques face limitations as low on-treatment image resolution reduces contrast, leading to difficulty isolating target volumes from surrounding OARs, in conjunction with target shape variability between treatments (46). For instance, a study achieved a 0.61 DSC

with the U-Net compared to 0.65 DSC with registration methods alone on CBCT in a population of 22 head and neck patients (47). As well, large U-Net infrastructures are often trained on large scale population datasets that do not reflect individual tumor response to day-to-day treatment during adaptive therapy. Patient-specific inputs are not learned by the model, leading to segmentation inaccuracies. Alternative infrastructures, like the ResNet model, a CNN designed to prevent overfitting inaccuracies, have been trialed to address these concerns. One such model was trained by Tie *et al.* on both low-field MR imaging pre-treatment and mid-treatment, fine-tuning the model on GTV masks mid-treatment to allow the model to learn GTV change from pre-treatment imaging (48). Their model, tested on 150 prostate cancer patients, yielded an average DSC of 0.61 for primary GTV and 0.86 for metastatic lymph nodes on treatment imaging.

Tie *et al.* also compared models trained with and without pre-treatment data, with clear model improvement shown when pre-treatment GTV information is included. Chen *et al.* furthered this work, training the similar DeepLabV3 ResNet model, but also fine-tuning the model for each individual patient on previous fractions for one-shot learning (30,49). On 26 prostate cancer patients, their model achieved a DSC of 0.93 ± 0.05 , compared to 0.79 ± 0.06 if using the ResNet model alone. These results suggest that capturing temporal deformations can help neural networks infer spatial evolution of disease.

Furthermore, to adapt U-Nets to specific patient contexts, “one-shot” training techniques have also been developed, where test data is not observed during training and the model is able to learn from one test data sample to make a reasonable prediction for the next test data sample (50,51). This allows active adaptation of models trained on a large dataset to specific patient contexts (the “test” data). Developments in one-shot learning include new techniques like test-time optimization (TTO), a method which individualizes pre-trained models to current-fraction images by iteratively refining the weights of the model in a traditional optimization manner (26). This method has demonstrated 5% improvements to DSC over other DL methods in head and neck cancer patients in the adaptive setting (26).

To further tailor AI segmentation algorithms to patient-specific changes during adaptive treatments, Kawula *et al.* studied the application of transfer learning for MR adaptive therapy (52). Transfer learning is described as the re-use and re-application of a model trained initially for one task to another. The study tested a trained three-dimensional (3D) U-Net on prostate cancer patients and applied transfer learning of the model to both a different facility and in patient-specific settings. There was no difference in DSC with transfer-learning of the algorithm to a different facility, with a DSC of 0.78 ± 0.07 in CTV contours compared to the base model evaluation of 0.73 ± 0.07 . Notably, however, transfer learning applied to individual patients improved CTV contour DSCs by 15% to 0.88 ± 0.05 . This demonstrates the role of transfer learning and the difficulty to apply models trained on the population level to individual patients.

Commercially, AI algorithms for adaptive CTV contouring have yielded some initially positive results. The commercial Varian Ethos system has been tested in gynecological sites. Adaptive plans required no major edits 92.3% of the time, and CTV segmentation ranged from a DSC of 0.76 (CTV-uterus) to 0.88 (CTV-cervix) (38).

Additionally, in head and neck cancer patients, Ethos was comparable or showed improved dosing to the OARs compared to traditional RTOG based plans in a study on 20 patients (53). However, the randomized DARTBOARD study in head and neck cancer showed that daily online adaptive radiation with 1–2 mm PTV margins decreased acute side effects but not xerostomia or other late side effects compared to standard radiation (54). They found that daily adaptive treatments took longer than 30 minutes and radiation oncologists were at the machine for more than 20 minutes per fraction.

While promising, GTV and CTV segmentation methods remain highly variable for adaptive treatments. As highlighted in De Biase *et al.*, the challenge in defining clinical targets is not only technical but conceptual (55). Integration of temporal information as well as personalized transfer learning and fine-tuning has been suggested to improve concordance of AI segmentations with expert delineations. Improvement to AI segmentations can also streamline treatment time, improving convenience for patients and clinicians. As such, ongoing development and clinical validation of these tools may benefit from seeking to capture additional context, providing models with more conceptual information to aid in segmentation.

AI decision support in ART

ART is most often guided by manual review of imaging changes, that may be subject to interobserver variability. To avoid interobserver subjectivity, a more objective and comprehensive approach known as Knowledge-Based Response-Adaptive Radiotherapy (KBR-ART) captures changes during online adaptive treatments and is a framework that allows for prediction. The KBR-ART framework consists of (I) pre-treatment planning using available knowledge; (II) updating predictive models with new knowledge during the therapy course; (III) leveraging the updated predictive models to adapt the treatment plan tailored to the patient’s response to treatment. KBR-ART could serve as a holistic, multi-modal framework to guide the design of AI-based ART, in which clinical, geometrical, and physiological changes all inform adaptations to treatment (56). Although it is argued that KBR-ART offers a more personalized approach, any subjectivity in treatment planning is compounded with each successive plan. Integrating other metrics beyond patient imaging into the decision-making process may also increase inter-physician variability.

As online ART moves treatment planning to the console in real time, there is a growing need for objective, data-driven decision support tools that can synthesize complex information and make recommendations for treatment. These tools include (I) dose-based decision support; (II) anatomy change classifiers; and (III) all-in-one adaptation workflow models.

Dose dependent decision support

To enable the dynamic nature of online ART, rapid and optimal plan generation is required. AI-driven dose prediction algorithms can predict tumor coverage and create OAR sparing plans quickly. These dose prediction algorithms allow clinicians to understand coverage at the console and toxicity risks to important OARs or re-treatment sites, guiding judgement and clinical decision making. Algorithms for dose calculation include the Halcyon Acuros XB (AXB) model and then Halcyon Anisotropic Analytical Algorithm model when using a Halcyon LINAC (57). These algorithms model photon and electron transport through heterogeneous media (bone, air, fluid, organ tissue) to calculate dose buildup. Systems like Varian Ethos apply the AXB algorithm to generate final dose calculations from an initial fluence optimization. However, these algorithms are designed for offline planning. To enable online adaptive decision support at the console, a quick and efficient algorithm to optimize treatment plans during a treatment session is paramount. Additionally, large models trained to generate dose plans are limited in their ability to personalize to daily adaptive motion with abnormal or altered anatomy that differs from patient to patient. AI tools provide potential solutions to address these problems.

One example of such a tool is the intelligent optimization algorithm (IOE) employed by the Ethos to generate plans within minutes at the console (53,58). The IOE sets up the optimization objective function by assigning appropriate weights to each optimization objective and continuously monitors the function during optimization. At the beginning of the planning process, the clinician will define the fractionation in the planning directive. Using clinical judgement from the physician, the planner will rank dosimetry goals into five categories for OARs and target volumes: (I) most important; (II) very important; (III) important; (IV) less important; and (V) report value only. The adaptive plan generated online at each fraction will then be optimized in terms of dose according to the ranking of the clinical goals. The IOE creates helper structures in

the background, using non-overlapping structures to resolve conflicting planning objectives and ring structures around targets to ensure dose drop-off at important sites. For example, one study found that applying the Ethos IOE in pelvic radiation, in settings where V90% of the bladder or bowel exceeded their clinical thresholds, adaptation reduced them by $13.1\% \pm 1.0\%$ and $6.5\% \pm 7.3\%$ (59).

While the Ethos system readjusts weights of the optimization objectives to account for patient's daily anatomical changes, achieving personalized, fast, high-quality plans is still difficult. Personalized dose models have been studied by applying transfer learning to fine-tune general AI models trained on a larger population on local patient cohorts. Fine-tuned patient-specific models have been clinically evaluated on head and neck patients to accurately predict dose. This model uses a U-Net architecture as a base model for dose prediction, and is then fine-tuned to each individual adaptive case using pre-treatment and previous treatment plans, customizing the algorithm to each patient. This model was tested on 23 adaptive fractions for 12 head and neck patients treated using the Ethos system (60). Comparing the Ethos system and the fine-tuned patient-specific model, the customized model reduced dose to high-impact OARs (constrictor, bilateral parotids, bilateral submandibular glands, oral cavity, esophagus, mandible, and larynx) with a maximum value of 5.47 Gy (61). Likewise, another study suggested that there is a statistically significant dose reduction in high-risk head and neck OARs using a fine-tuned, patient-specific U-Net model compared to the same U-Net model trained on 33 patients without fine-tuning (62). Transfer learning has also been applied to improve dose prediction in prostate cancer patients. Kandalan *et al.* fine-tuned a U-Net on a dataset of 108 patients, achieving dose-volume histogram predictions within 1.6% to ground truth plans (63).

AI demonstrates potential to aid in decision making through dose prediction, helping balance the risk of toxicity for patients at each fraction. AI has been applied to make dose prediction more efficient and feasible for daily online adaptive therapies and can also be applied to personalize dose prediction tailored to patient-specific anatomical changes between fractions. Future work is needed however to correlate dose predictions with toxicity outcomes. While AI has been utilized to predict long-term toxicity in conventional radiotherapy, this work has not been well-studied for daily adaptive treatment responses or in the long-term setting for patients receiving adaptive treatment.

Anatomical change classifiers

Anatomical changes throughout the course of radiotherapy, including tumor regression, weight loss, and daily variation in organ shape or motion, is a primary motivation for ART. Prediction of anatomical change before treatment can be useful to decide good candidates who may benefit from ART. As well, prediction of anatomical change during treatment can inform potential toxicities, plan change, and future daily treatment planning.

Current prediction of patients suitable for ART is based on clinical judgment and daily reimaging, as well as proximity to anatomical structures like the bowel and bladder which contribute to interfraction motion. However, AI tools are emerging that support decision making before or during adaptive treatments to recommend when plan adaptation is necessary and the patients that can benefit the most from these techniques. An early example of this includes a machine learning model that trained a Bayesian classifier on features extracted from 119 reference and daily on treatment MRI images of patients undergoing pancreatic cancer radiotherapy (64). When tested on a cohort of 24 patients, this model was able to predict correctly when online adaptation was necessary for subsequent treatment in patients 98% of the time. Similar work was done in head and neck treatment using a support vector machine, machine learning classifier model trained on relevant radiomic features to achieve a 0.9 classification performance accuracy between patients needing adaptive treatments and those who did not (65). A support vector machine was also trained on relevant radiomic features extracted from 671 daily CBCT images across 24 patients with cervical cancer, with an area under the curve of 0.884 in predicting adaptation. Notably, this study also evaluated model performance *vs.* physician consensus and found that the model outperformed physicians with an accuracy of 0.855 *vs.* 0.795 (66).

Although simple machine learning techniques have demonstrated success in small datasets, large-scale deep learning techniques have also been implemented for decision support. Convolutional long-short term memory (LSTMs) networks have been studied in their ability to predict for interfraction changes in anatomy. LSTMs, an architecture originally designed for use in video processing, is trained on image sequences and can generate and predict the next image expected according to the previous sequence. Lee *et al.* applied a pre-trained LSTM to predict weekly GTV change in 16 patients receiving ART to lung or esophageal primary cancers (67). They found that by using just daily adaptive imaging, they were able to achieve a DSC

concordance of 0.71 to 0.83 for weekly predictions. Dose plans generated by weekly predicted contours maintained 60 Gy tumor coverage in all plans.

While current data is limited to single-center trials on fewer patient cohorts, this work signifies that machine learning and deep learning techniques can potentially predict the need for adaptation. However, a major limitation in this work is that pre-trained models cannot account for individual patient responses. This potentially accounts for the better performance of simpler machine learning algorithms compared to large-scale deep learning networks trained on large population datasets. Transfer learning or fine-tuning larger pre-trained LSTM models may be necessary for clinical translation and to improve concordance.

Adaptation workflows

Recent work has integrated of anatomical change classifiers, auto-segmentation algorithms, and dose planning neural networks, to create holistic ART workflows.

A multi-disciplinary team from Moffitt Cancer Centre and the University of Michigan developed the first interactive software dedicated to creating a more unified approach to KBR-ART decision making (68). Their tool, Adaptive Radiotherapy Clinical Decision Support (ARClIDS), is AI-based with two main components. The first component of ARClIDS, the Artificial RT Environment (ARTE), utilizes a Markov process, a dynamic statistical model of sequential decision-making with uncertain outcomes, to predict treatment outcomes based on an input plan's dosimetry metrics. Using Markov modeling, uncertainties such as patient behaviour/adherence, effectiveness of treatment, and side effects can all be collectively modelled based by assigning each variable a probability of its occurrence based on *a priori* knowledge. The two outcomes estimated by ARTE are probabilities of tumour control and normal tissue complication, generated as a function of daily radiation doses when given patient information such as dosimetric, clinical, radiomics, and genomics parameters (68). This is integrated for decision making to decide whether adaptive treatments are necessary in the next fraction. The second component, optimal decision making (ODM), is trained to recommend the optimal daily dose adjustments for maximizing tumour control and minimizing toxicities based on data provided by ARTE. The ODM considers the outcomes predicted by ARTE against their associated uncertainties to provide a recommendation of a daily dose to optimize clinical outcomes.

ARClIDS was retrospectively evaluated on datasets of 117

non-small cell lung cancer patients receiving ART and 292 hepatocellular carcinoma patients receiving adaptive SBRT (69,70). The ARCLiDS system achieved a mean difference of 0.61 ± 0.03 Gy per fraction to the CTV in lung cancer patients and 2.96 ± 0.42 Gy per fraction for hepatocellular carcinoma patients. The system also improved up to 74% of treatment decisions that led to toxicities and complications. Although there are no commercially available softwares using this at this time, creation of systems like ARCLiDS and further formal testing in a clinical cohort can potentially contribute to improvements in clinical decision making and allow for streamlined integration of AI into online adaptive techniques.

Outside of ARCLiDS, other KBR-ART adaptation workflow systems have been clinically trialed in head and neck cancer patients. Gros *et al.* trialed the “RTapp”, which estimated daily dose distribution and cumulative doses in 27 head and neck cancer patients receiving adaptive treatments. Analyzing dosimetric parameter trends against user-specified thresholds, warnings to trigger adaptation of plan could be foreseen up to 4 fractions ahead. The difference between predicted and calculated PTV V95 and parotid Dmean was up to 7.6% in the patient cohort, and demonstrates ability for AI decision making to impact adaptive workflows. While some of this work has been piloted in single-centre clinical workflows, further testing on larger cohorts between institutions is needed to demonstrate further utility (71,72).

Conclusions

AI is reshaping ART by automating labor- and time-intensive steps, such as synthetic CT generation, auto-contouring, image registration, dose prediction/optimization, and decision support, so that adaptations can be made reliably within clinical time constraints. Early deployments of AI-enabled ART, such as Varian’s Ethos, have demonstrated promising non-inferior performance compared to manual ART.

Broad clinical adoption first requires multi-institutional, prospective validation with patient-centered endpoints. With thoughtful implementation, AI-driven ART can move towards realizing a workflow that personalizes dose and target definition over the course of radiotherapy while preserving safety through human-in-the-loop verification.

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Footnote

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